

Research Notes: 2026 Agent Memory API and Data Model Best Practices

1. Memory Architecture Trends (July 2026)

- **Two-Tier/Layered Architecture:** Standard practice involves a fast, transient "Working Memory" (LLM context window) and a persistent, searchable "Long-term Memory" [10] [11].
- **Hybrid Storage:** Combining SQLite for structured data (user preferences, session metadata) with vector databases for semantic search (unstructured memories) [7] [8].
- **Local-First/Offline-First:** Strong emphasis on local-first memory engines (e.g., "SQLite for AI Agents") that provide privacy and low latency for local LLMs [11].
- **Agentic Memory:** Allowing the LLM to decide what to remember and how to organize it, rather than manual modeling [15].

2. Data Model Best Practices (July 2026)

- **Memory Object Schema:**
 - `id` : Unique identifier (UUID).
 - `content` : The memory text.
 - `type` : Categorization (e.g., `preference` , `fact` , `history`).
 - `importance` : A weight or score indicating memory relevance.
 - `timestamp` : When the memory was created or last updated [12].
 - `metadata` : JSON object for contextual data (session ID, source).
- **Session Schema:**
 - `id` : Unique session identifier.
 - `user_id` : Link to user profile.
 - `summary` : Periodically updated summary of the session to save context.
 - `created_at` : Session start time.

3. API Design Trends (July 2026)

- **Memory Lifecycle Management:** Endpoints for creating, retrieving, updating, and deleting (CRUD) memories [1] [2].

- **Semantic Search Endpoints:** Allowing agents to query their memory based on meaning rather than just keywords [9] [11].
- **Context Injection APIs:** Endpoints that provide a formatted "memory block" to be injected into the LLM prompt.

References

1. [REST API for AI Models Explained: 2026 Guide](#)
2. [API design best practices guide \(March 2026\)](#)
3. [AI Agent Memory Architecture Guide: Three Paths to the Right ...](#)
4. [AI Agent Memory Systems Compared: RAG vs Local SQLite vs ...](#)
5. [GitHub - sqliteai/sqlite-memory: Markdown based AI agent memory ...](#)
6. [How to Build and Operate AI Agent Memory in 2026](#)
7. [I built "SQLite for AI Agents" A local-first memory engine with hybrid ...](#)
8. [A Practical Guide to Memory for Autonomous LLM Agents](#)
9. [How to build your own long-term Agentic Memory System for LLMs](#)